

The key reasons why blockchain technology is becoming popular in security aware applications are:

- Resilience
- Time reduction
- Reliability
- Unchangeable transactions
- Fraud prevention
- Security
- Transparency
- Collaboration
- Decentralised

The flow of processes in the blockchain environment is shown in Figure 2. Each and every transaction is validated with the implementation of advanced protocols, and cannot be cracked because of the smart contracts integrated in the blockchain network.

Case studies and applications of blockchains

As per reports from Statista, global spending on blockchains moved to around US\$ 3 billion in 2019. Blockchain technology is not confined to e-governance or e-commerce, but also advancing in the domain of agriculture. The use of the blockchain in the agriculture segment amounted to a global turnover of over US\$ 40 million in 2018.

The United Arab Emirates (UAE) is working on the development of a blockchain based city with fully featured blockchain implementations. Smart Dubai is proposed to be the first city that will be fully powered with blockchain technology. The research reports claim that UAE will save more than US\$ 3 billion by deploying the blockchain for multiple sectors.

Krisflyer Singapore Airlines is adopting blockchain technology with the integration of a digital wallet. A loyalty programme and payment have also been integrated with this technology.

Singapore has been very active in the implementation of blockchain technology for multiple applications including

Table 1: High performance technologies for blockchain programming

Technology/Platform	URL
Corda	https://www.corda.net/
Hyperledger	https://www.hyperledger.org/
BigChainDB	https://www.bigchaindb.com/
HydraChain	https://github.com/HydraChain/hydrachain
OpenChain	https://www.openchain.org/
Eris	https://erisindustries.com/
MultiChain	http://www.multichain.com/
IOTA	http://iotatoken.com/
Chain	https://chain.com/
Ethereum	https://www.ethereum.org/
IBM Bluemix Blockchain	https://console.ng.bluemix.net

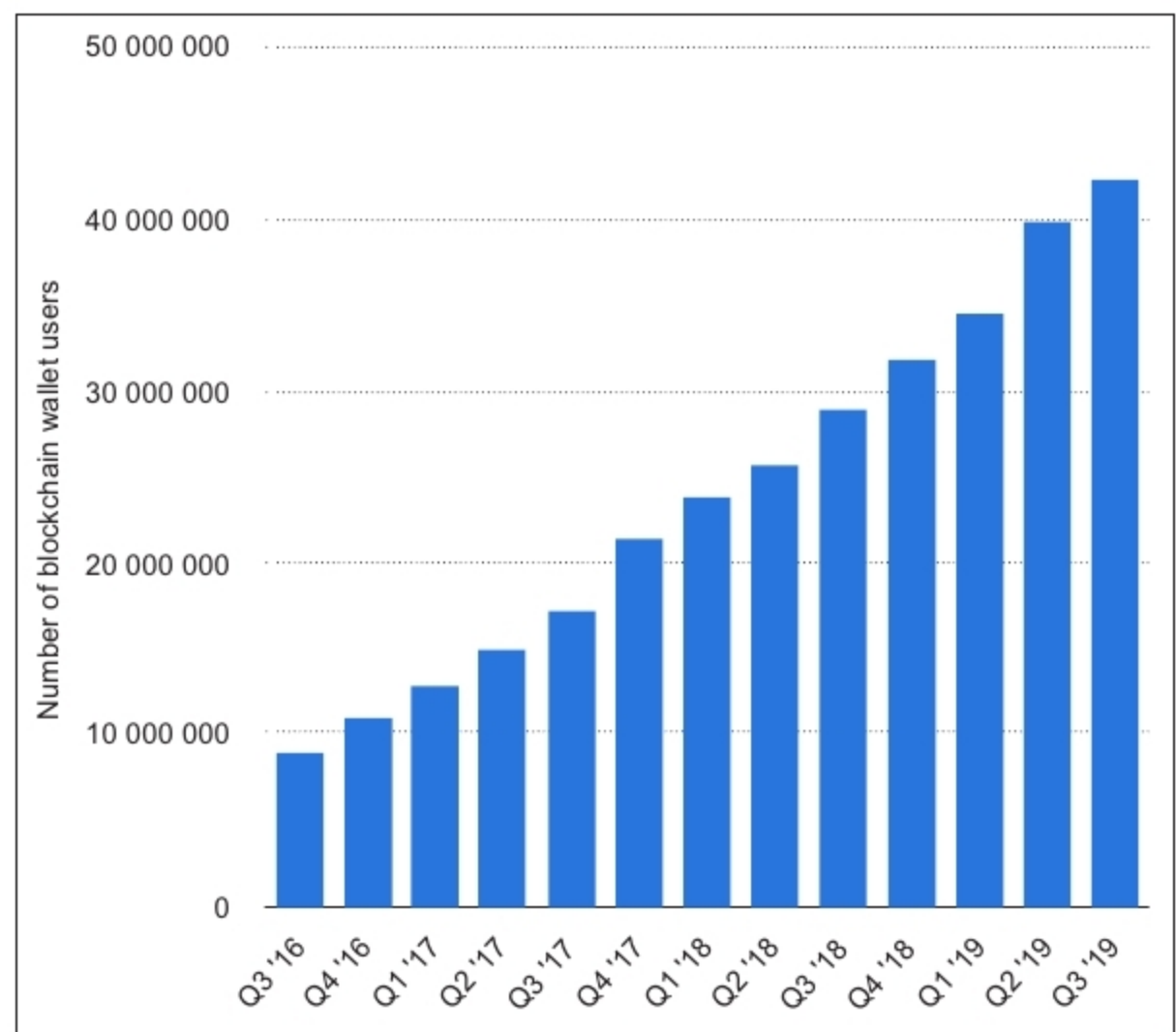


Figure 1: Users of blockchain wallets in the years 2016 to 2019

education, e-governance, the food industry, healthcare and many others.

In the real estate sector, Averspace is known to be the first portal that is blockchain enabled for the selling and renting of properties. The transactions are deployed in a secured way with digital smart contracts.

Free and open source tools for blockchain programming

A number of platforms and technologies are available for blockchain programming. The key factors to consider when choosing the appropriate platform for blockchain programming include the type of network, the